

A Strategy to Improve the Convergence of the Fixed-Point Method for Nonlinear Eddy Current Problems

Stefan Außerhofer, Oszkár Bíró, and Kurt Preis

Institute of Fundamentals and Theory in Electrical Engineering, Graz University of Technology, 8010 Graz, Austria

The fixed-point method together with the Fourier decomposition of time dependent quantities can be used to determine the steady state solution of sinusoidally driven electrodynamic problems in the presence of nonlinear materials. The advantage of this method is to obtain the steady state immediately without having to calculate transient processes. However, the convergence rate of the method is heavily influenced by the so called fixed-point permeability. An appropriate choice of this parameter can speed up the convergence whereas an unsuitable value leads the method to divergence. The aim of this paper is to present a strategy to choose the value for the fixed-point permeability in order to enhance the convergence rate. The result is illustrated by two 2-D examples.

Index Terms—Eddy currents, frequency domain analysis, nonlinearities, numerical analysis.

I. INTRODUCTION

THE nonlinear behavior of ferromagnetic materials can usually not be neglected in electrodynamic problems. Because of these nonlinearities, such problems are typically solved in the time domain. Therefore, the time derivatives are approximated by finite difference quotients and the field distribution at a certain time step is calculated with the aid of the solution at previous time steps. With this method, the transient process has to be stepped through even if steady state is required only. If the transients are slowly decaying, a lot of calculations have to be made to arrive at steady state which uses a lot of computational effort.

The fixed-point method is a promising technique to determine the steady state without calculating transient processes (see e.g., [1] and [2]). However, the convergence rate of the method depends on the fixed-point permeability, μ_{FP} . In the static case, there exists a factor μ_{FP} unconditionally ensuring convergence (see [3]), but using this value the convergence is usually slow. To speed up the convergence rate, it is necessary to have a strategy for choosing an appropriate μ_{FP} which leaves the fixed-point method convergent. In this paper, a method is proposed to accelerate the iteration process. The method was tested on two two-dimensional examples.

II. FIXED-POINT METHOD AND FOURIER DECOMPOSITION

The nonlinear behavior of the ferromagnetic material is given by the BH-curve. The relationship between the flux density $\mathbf{B}$ and the magnetic field intensity $\mathbf{H}$ is separated into a linear and a nonlinear term:

$$\mathbf{H}(\mathbf{B}) = \frac{1}{\mu_{\text{FP}}} \mathbf{B} - \mathbf{M}_{\text{FP}}(\mathbf{B}). \quad (1)$$

The term $\mathbf{M}_{\text{FP}}$ in (1) is a magnetization-like quantity which describes the nonlinearity and therefore depends on the flux density $\mathbf{B}$. The constant factor μ_{FP} of the linear term is called the fixed-point permeability.

Relation (1) together with Ampere's law yields

$$\frac{1}{\mu_{\text{FP}}} \text{curl } \mathbf{B} = \mathbf{J} + \text{curl } \mathbf{M}_{\text{FP}}. \quad (2)$$

The differential equation (2) can be seen as Ampere's law in the linear case for a material with a constant permeability μ_{FP} and driven by the excitation $\mathbf{J} + \text{curl } \mathbf{M}_{\text{FP}}$, whereas $\mathbf{J}$ denotes the current density.

If the ferromagnetic material is conductive as well, additional equations are necessary

$$\begin{aligned} \text{curl } \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \mathbf{J} &= \sigma \mathbf{E}. \end{aligned} \quad (3)$$

$\mathbf{E}$ denotes the electric field intensity and σ is the conductivity of the material.

It is assumed that the excitation is sinusoidal with the angular frequency ω_0 . Due to the nonlinearity of the material, the time dependent quantities $\mathbf{B}$, $\mathbf{J}$, $\mathbf{M}_{\text{FP}}$, and $\mathbf{E}$ are no longer sinusoidal but they are still periodic. Therefore, they can be approximated by Fourier series with a finite number of harmonics. The approximate representation of a time dependent quantity $\mathbf{X}(t)$ is given by

$$\mathbf{X}(t) = \text{Re} \left\{ \sum_{k=1}^M \mathbf{X}_k e^{jk\omega_0 t} \right\}. \quad (4)$$

In (4), the amplitude of $\mathbf{X}(t)$ at the angular frequency $k\omega_0$ is denoted by $\mathbf{X}_k$ and M is the number of harmonics. The Fourier representations for the time domain quantities in (2) and (3) are easily obtained by replacing $\mathbf{X}$ by $\mathbf{B}$, $\mathbf{J}$, $\mathbf{M}_{\text{FP}}$, or $\mathbf{E}$.

Since the exponential functions $\exp(jk\omega_0 t)$ are orthogonal for different values of k , applying (4) to (2) and (3) results in a number of decoupled systems of differential equations for each harmonic component:

$$\begin{aligned} \frac{1}{\mu_{\text{FP}}} \text{curl } \mathbf{B}_k &= \mathbf{J}_k + \text{curl } \mathbf{M}_{\text{FP},k} \\ \text{curl } \mathbf{E}_k &= -jk\omega_0 \mathbf{B}_k \\ \mathbf{J}_k &= \sigma \mathbf{E}_k \\ \text{with } k &= 1, \dots, M. \end{aligned} \quad (5)$$

Due to the Fourier representation of the time dependent quantities, the time derivative in (3) has been replaced in (5) by a

multiplication with $jk\omega_0$. Notice that the solution of (5) is a function of the magnetization $\mathbf{M}_{\text{FP}}$.

The iterative process starts with an arbitrary value for $\mathbf{M}_{\text{FP}}$ (typically zero). Given a certain value for $\mathbf{M}_{\text{FP}}$, the field density $\mathbf{B}$ in the frequency domain can be obtained by solving (5). Since the frequency components are known, the associated time dependent quantity $\mathbf{B}(t)$ is given by (4). With $\mathbf{B}(t)$ and using relation (1), a new $\mathbf{M}_{\text{FP}}(t)$ can be determined from the given BH-curve. This leads to a new right-hand side of the first differential equation in (5) and hence to a new flux density $\mathbf{B}$. This algorithm is repeated, until a certain criterion is fulfilled.

III. STRATEGY FOR FIXED-POINT PERMEABILITY

A value $\mu_{\text{FP}}(P) < 2 \cdot \mu_{\min}(P)$ for the fixed-point permeability in a point P ensures convergence in the static case (see [3] and [4]). Here, $\mu_{\min}$ is the minimum slope of the BH-curve, which is usually equal to the permeability μ_0 of vacuum. However, numerical investigations show (see Section IV) that higher values for μ_{FP} accelerate convergence, but if μ_{FP} exceeds a certain value, the fixed-point method diverges. The aim is to choose a μ_{FP} higher than μ_0 which leaves the method convergent.

If, in the dynamic case, the distribution of the flux density $\mathbf{B}$ in the nonlinear domain were known, the maximum value of $\mathbf{B}$ in a point P within a period can be determined by

$$\mathbf{B}_{\max}(P) = \max_t \{ \mathbf{B}(P, t) \}, \quad 0 \leq t < \frac{2\pi}{\omega_0}. \quad (6)$$

The corresponding permeability for the maximum of the flux density at this point is given by

$$\mu^*(P) = \frac{|\mathbf{B}_{\max}|}{|\mathbf{H}(\mathbf{B}_{\max})|}. \quad (7)$$

If this value is chosen to be the fixed-point permeability μ_{FP} at the point P , and if it is further assumed that $\mathbf{M}_{\text{FP}}$ is also known, the value of $\mathbf{M}_{\text{FP}}$ must be zero for the time where the maximum flux density occurs [cf. (1)]. For times, where the flux density is lower than the maximum value, $\mathbf{M}_{\text{FP}}$ is not zero but bounded. An overall upper bound for $\mathbf{M}_{\text{FP}}$ can be obtained, if μ_{FP} is set to the permeability of vacuum.

For the iterative algorithm, the fixed-point permeability μ_{FP} in each integration point is replaced by (7) at each iteration step. It is assumed that this choice will improve the convergence rate, since the value determined by (7) is, in general, higher than μ_0 . To determine the magnetization $\mathbf{M}_{\text{FP}}$, this new value for μ_{FP} is used in (1). This ensures that $\mathbf{M}_{\text{FP}}$ is bounded which should guarantee stability.

The iterative algorithm is as follows.

- 1) Choose an initial value for μ_{FP} .
- 2) Set $\mathbf{M}_{\text{FP}}$ to zero.
- 3) Solve (5) to calculate $\mathbf{B}$.
- 4) Determine

$$\mathbf{B}_{\max} = \max_t \{ \mathbf{B}(t) \} \\ \mu^* = \frac{|\mathbf{B}_{\max}|}{|\mathbf{H}(\mathbf{B}_{\max})|}$$

in each integration point.

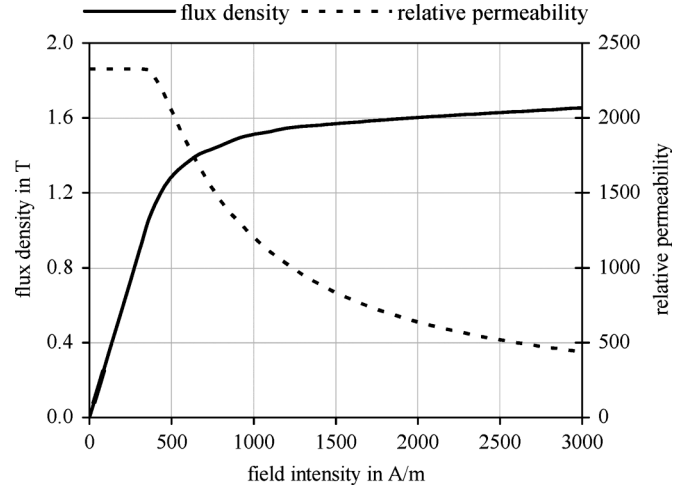


Fig. 1. BH-curve and relative permeability of ferromagnetic material.

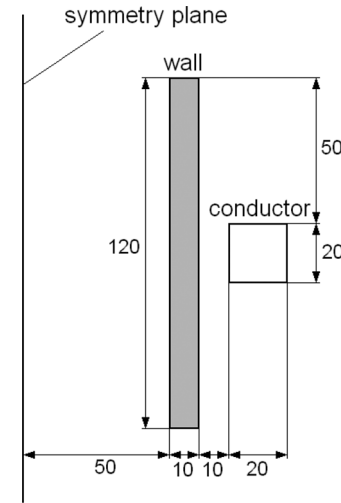


Fig. 2. Aluminum conductor with ferromagnetic wall.

- 5) Set $\mu_{\text{FP}} = \mu^*$ in each integration point and calculate

$$\mathbf{M}_{\text{FP}} = \frac{1}{\mu_{\text{FP}}} \mathbf{B} - \mathbf{H}(\mathbf{B}).$$

- 6) Update the right-hand side of (5).
- 7) Repeat steps 3–6 until a certain criterion is fulfilled.

IV. NUMERICAL EXAMPLE

The algorithm described above was tested on two simple two-dimensional problems. The BH-curve of the ferromagnetic material, used for both examples, is given in Fig. 1. The maximum value of the relative permeability for the material is about 2330.

A. Aluminum Conductor With Ferromagnetic Wall

The first example shown in Fig. 2, consists of an aluminum conductor with a conductivity of $\sigma = 3.5 \cdot 10^7$ S/m. The conductor is driven by a sinusoidal voltage of 0.66 V per meter and the excitation frequency is 50 Hz. The ferromagnetic wall has a conductivity of $\sigma = 10^7$ S/m and the BH-curve of the material is given in Fig. 1.

The calculations have been carried out for constant values of μ_{FP} and with the present method too. The stopping criteria are

TABLE I
RESULTS OF THE FIRST EXAMPLE

method	$\mu_{r,FP}$			number of iterations
	initial value	min	max	
constant μ_{FP}	50	-	-	91
	100	-	-	50
	300	-	-	26
	600	-	-	24
μ_{FP} adapted by (7)	50	316	2326	22
	600	316	2326	19
	2330	318	2326	21

0.1% for the mean relative variation and 1% for the maximum relative variation of the permeability $\mu(t) = |\mathbf{B}(t)|/|\mathbf{H}(t)|$ in the integrations points. Nine harmonics have been used for the Fourier decomposition to achieve accurate results.

Table I show the results of the calculation, where $\mu_{r,FP}$ denotes the relative fixed-point permeability. The first four rows are the results if μ_{FP} is constant. In this case, the number of iterations decreases if the relative permeability $\mu_{r,FP}$ increases. The minimum number of iterations, where the method is still stable, is obtained for $\mu_{r,FP} = 600$. If $\mu_{r,FP}$ is further increased to 700, the method fails to converge.

For the present method where μ_{FP} is updated by (7), the number of iterations is about the same for different initial values of $\mu_{r,FP}$. The number of iterations is as high as in the case with the maximum value of $\mu_{r,FP}$ constant. An important point is that the present method also converges, if the starting value for $\mu_{r,FP}$ is higher than 700 (see last row in Table I). The minimum and maximum values of the relative fixed-point permeability in the integration points found by the present method are also given in Table I. The values vary between 300 and the maximum value of the relative permeability of the BH-curve. For the case of $\mu_{r,FP}$ fixed in each integration point to the value found by the adaptive method, only 19 iterations are necessary to fulfill the stopping criterion.

The results are also shown in Fig. 3 where the evolution of the maximum variation of the permeability is given. It can be seen that the rate of convergence for the present method (curve b) is higher than for a constant $\mu_{r,FP}$ of 50 or 100 (curves a1 and a2). Just for comparison, curve c shows the case where μ_{FP} is constant but fixed to the values obtained by the present method. The convergence rate of curve c is about the same as for curve b (present method).

B. Copper Conductor With Ferromagnetic Screen

The second example consists of a copper conductor surrounded by a ferromagnetic screen having an air-gap. The BH-curve of the ferromagnetic material is the same as in the first example (see Fig. 1). The conductivities are $\sigma = 5.7 \cdot 10^7$ S/m for the copper and $\sigma = 10^6$ S/m for the ferromagnetic material, respectively. A voltage of 1.4 V per meter is applied to the copper conductor and the frequency of the excitation is again 50 Hz. The dimensions of the example are given in Fig. 4. For this example, the transient processes decay much slower than in the first example and, therefore, more periods are needed to arrive at steady state if time stepping is employed (see [1] and [2]).

Nine harmonics are used for the Fourier representation of time dependent quantities. The stopping criterion for the non-

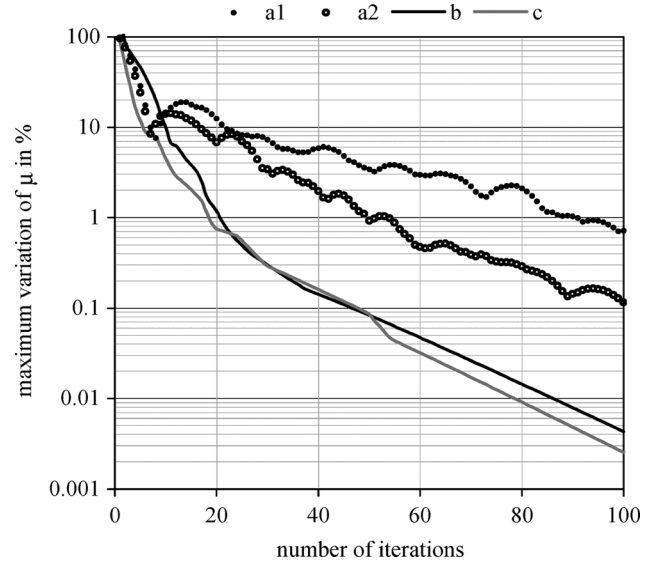


Fig. 3. Maximum variation of the permeability; a1: $\mu_{r,FP} = 50$; a2: $\mu_{r,FP} = 100$; b: $\mu_{r,FP}$ adapted; c: $\mu_{r,FP}$ fixed to the values obtained by the present method.

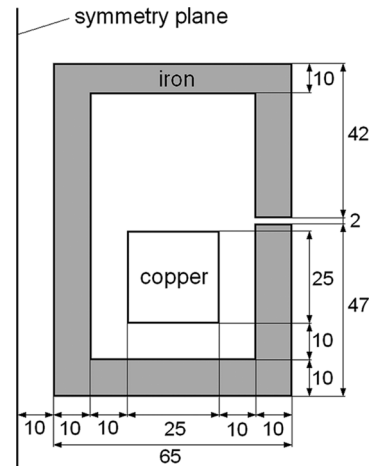


Fig. 4. Copper conductor with ferromagnetic screen.

TABLE II
RESULTS OF THE SECOND EXAMPLE

method	$\mu_{r,FP}$			number of iterations
	initial value	min	max	
constant μ_{FP}	50	-	-	120
	100	-	-	67
	300	-	-	54
	500	-	-	54
μ_{FP} adapted by (7)	50	35	2326	53
	500	35	2326	53
	2330	35	2326	53

linear iteration is set to 0.1% for the mean, and 1% for the maximum variation of the permeability $\mu(t) = |\mathbf{B}(t)|/|\mathbf{H}(t)|$.

Table II shows that the higher $\mu_{r,FP}$ is, the less iterations are needed in the case of a constant fixed-point permeability. The upper bound for $\mu_{r,FP}$, for which the method is still stable, is $\mu_{r,FP} = 500$. If, in contrast, the fixed-point permeability is determined by (7), the initial value for $\mu_{r,FP}$ can be chosen higher than 500 without becoming divergent (last row of Table II). In

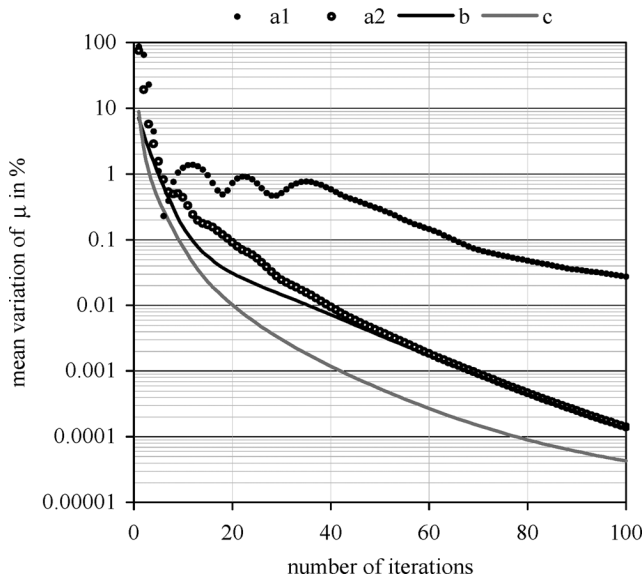


Fig. 5. Mean variation of the permeability; a1: $\mu_{r,FP} = 50$; a2: $\mu_{r,FP} = 300$; b: $\mu_{r,FP}$ adapted; c: $\mu_{r,FP}$ fixed to values from the present method.

this case, the number of iterations is about the same as in the case of a constant permeability with the minimum number of iterations. When μ_{FP} is fixed to the values obtained by the adaptive method, the number of iteration is only 34.

The evolution of the mean variation of the permeability is shown in Fig. 5. The curves a1 and a2 show the behavior for different values of a constant μ_{FP} , whereas curve b show the evolution of the mean variance when μ_{FP} is updated by (7). Curve c shows the result if μ_{FP} is fixed to the values taken from the adaptive process. So, if the values for μ_{FP} would have been known a priori, the convergence is the best over all.

V. CONCLUSION

The simple numerical examples show that the method of choosing μ_{FP} by (7) improves the convergence in general. The values for the fixed-point permeability determined by this method seem to be good starting values as can be seen from curve c in Fig. 4 (second example). The method is also stable for high values of the starting values μ_{FP} and, as can be seen from the calculations, the convergence rate is nearly independent of the choice of the starting value. Therefore, it is not necessary to have knowledge about a suitable initial value for μ_{FP} . It can also be seen that the number of iterations in the case of a fixed permeability is higher or equal to the number of iterations obtained by the present method.

ACKNOWLEDGMENT

This work was supported in part by the Austrian Science Fund (FWF) under the project number P18479-N07.

REFERENCES

- [1] O. Bíró and K. Preis, "An efficient time-domain method for nonlinear periodic eddy current problems," *IEEE Trans. Magn.*, vol. 42, no. 4, pp. 695–698, Apr. 2006.
- [2] S. Ausserhofer, O. Bíró, and K. Preis, "An efficient harmonic balance method for nonlinear eddy current problems," *IEEE Trans. Magn.*, vol. 43, no. 4, pp. 1229–1232, Apr. 2007.
- [3] F. I. Hanttila, G. Preda, and M. Vasiliu, "Polarization method for static fields," *IEEE Trans. Magn.*, vol. 36, no. 4, pp. 672–675, Jul. 2000.
- [4] F. I. Hanttila, "Mathematical models of the relation between $\mathbf{B}$ and $\mathbf{H}$ for non-linear media," *Rev. Roum. Sci. Techn.-Electrotechn. et Energ.*, vol. 19, no. 3, pp. 429–448, 1974.
- [5] M. Chiampi, C. Ragusa, and M. Repetto, "Strategies for accelerating convergence in nonlinear fixed point method solutions," in *Proc. 7th Int. IGTE Symp.*, Graz, Austria, Sep. 23–25, 1996, pp. 245–250.

Manuscript received June 24, 2007. Corresponding author: S. Außerhofer (e-mail: stefan.ausserhofer@tugraz.at).